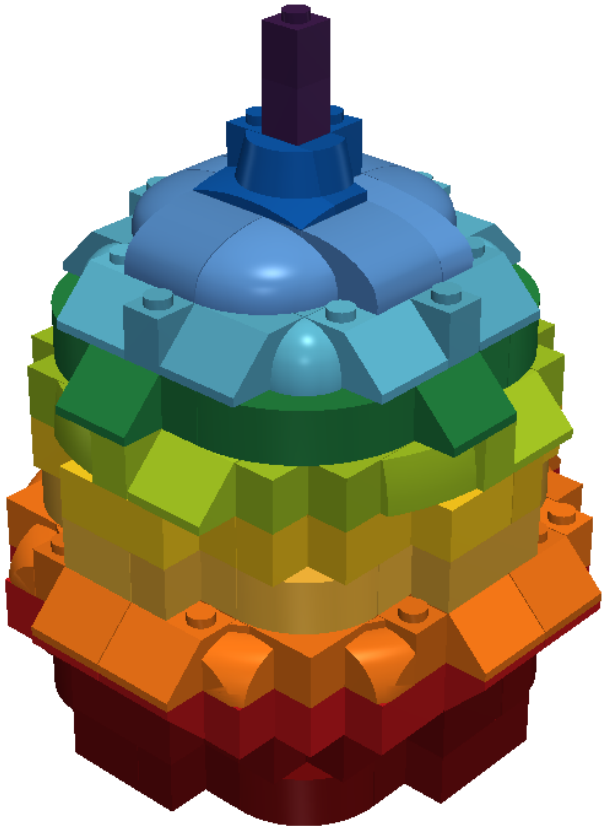
capsule

Building manual – Legolize it



Parts	191, 58 shaped, 12 special
Layers	13 (bricks, 9.6 mm)
Size	104 × 104 × 134.4 mm
Grid	13 × 13 × 14 cells, pitch 8 mm
Stud play	0.15 mm (wall 1.35, tube 6.2137, socket 5.1 mm)
Holds together	yes

Parts

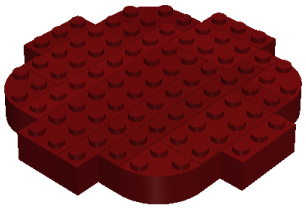
Part	Colour	Count
2x8	bright light orange	1
2x8	dark red	4
2x8	lime	1
2x8	red	3
2x8	yellow	3
2x6	green	1
2x6	lime	1
2x6	orange	3
1x8	azure	1
1x8	bright light orange	1
1x8	dark red	4
1x8	lime	3
1x8	orange	2
1x8	red	3
1x8	yellow	1
2x4	bright light orange	2
2x4	dark red	2
1x6	bright light orange	2
1x6	dark red	1
1x6	green	1
1x6	orange	1
1x6	yellow	2
2x3	azure	1
2x3	bright light orange	1
2x3	dark red	2
2x3	green	2
2x3	medium blue	1
2x3	orange	3
2x3	red	1
2x3	yellow	1
1x4	azure	1
1x4	dark red	1
1x4	lime	2
1x4	red	1
2x2	azure	3
1x3	bright light orange	5
1x3	dark red	7
1x3	green	1
1x3	lime	2
1x3	orange	3
1x3	red	1
1x3	yellow	3
1x2	azure	1
1x2	bright light orange	1
1x2	dark red	4
1x2	green	1
1x2	lime	1
1x2	medium blue	1
1x2	red	2
1x2	yellow	2
1x1	azure	1

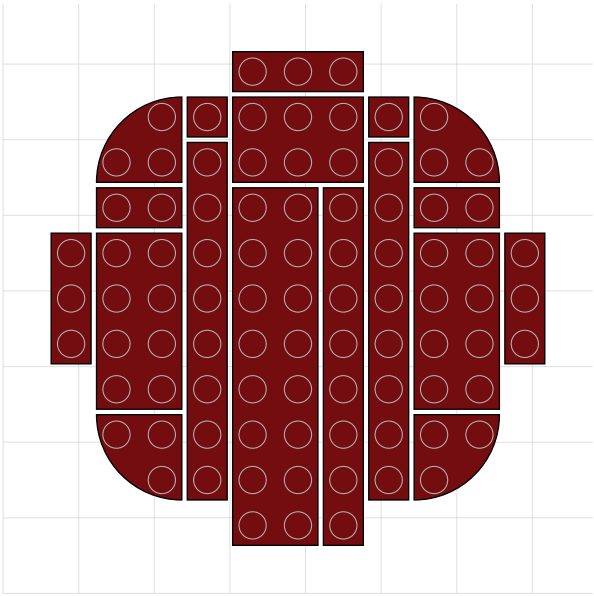
1x1	bright light orange	2
1x1	dark purple	2
1x1	dark red	3
1x1	green	2
1x1	lime	5
1x1	orange	2
1x1	red	4
1x1	yellow	2
round corner 3x3	green	4
curve 2x2	medium blue	3
dome corner 2x2	medium blue	4
inverted slope 2x2	red	8
round corner 2x2	blue	1
round corner 2x2	bright light orange	4
round corner 2x2	dark red	4
slope 2x2	azure	8
slope 2x2	green	4
slope 2x2	lime	4
slope 2x2	orange	8
curve 3x1	medium blue	1
special 1x3	blue	1
curve 2x1	medium blue	1
special 1x2	lime	4
dome corner 1x1	azure	4
special 1x1	blue	1
special 1x1	orange	6
all		191

How to read the steps

Each step is one layer. The picture shows the model so far, the new layer bright. The plan shows the layer from above: the layer below in grey, every part in its colour, circles for studs. An arrow points down the fall of a slope or curved slope (dashed: an inverted one, falling underneath). Quarter circles are round or dome corners, full circles round bricks. Hatched parts marked **S***n* are special pieces: custom parts with the input's exact shape, printed from the STL stack (`bricks/special_NNNNN.stl`).

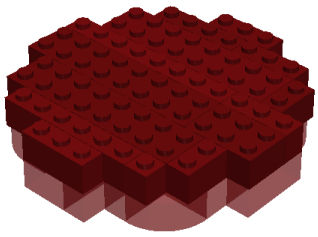
Step 1 (layer 0, z = 0 mm, 18 parts)

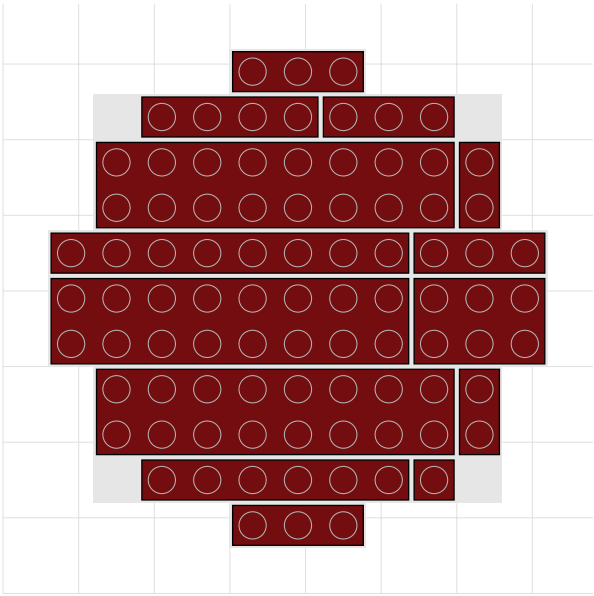




Part	Colour	Count
2x8	dark red	1
1x8	dark red	3
2x4	dark red	2
2x3	dark red	1
1x3	dark red	3
1x2	dark red	2
1x1	dark red	2
round corner 2x2	dark red	4

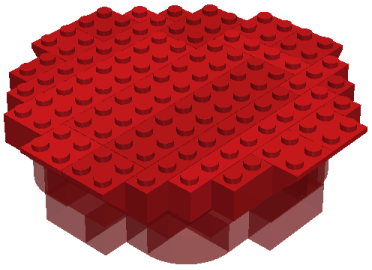
Step 2 (layer 1, z = 9.6 mm, 14 parts)

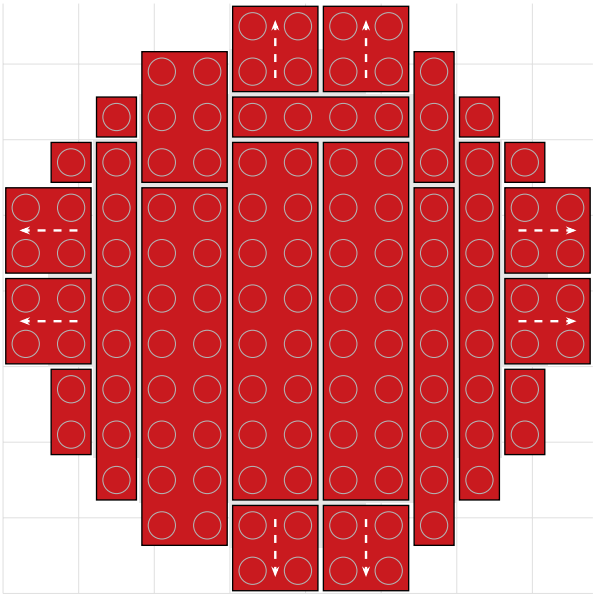




Part	Colour	Count
2x8	dark red	3
1x8	dark red	1
1x6	dark red	1
2x3	dark red	1
1x4	dark red	1
1x3	dark red	4
1x2	dark red	2
1x1	dark red	1

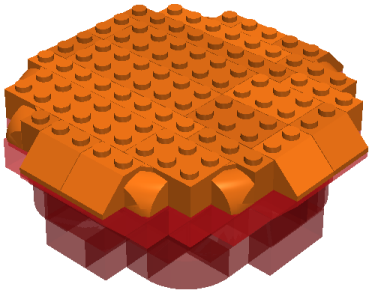
Step 3 (layer 2, z = 19.2 mm, 23 parts)

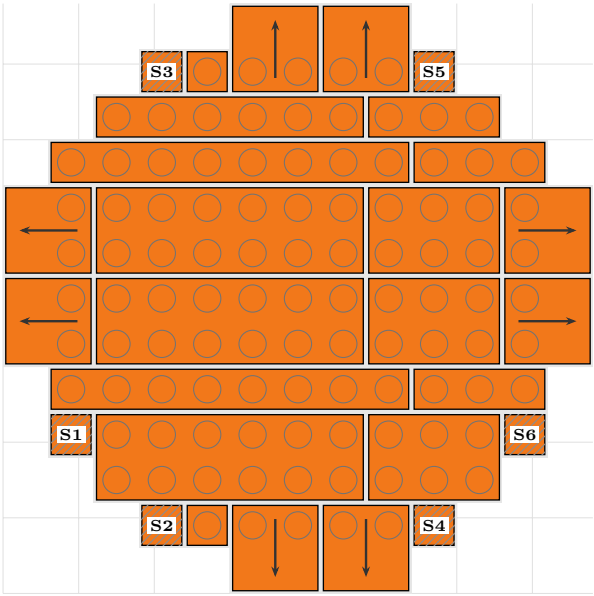




Part	Colour	Count
2x8	red	3
1x8	red	3
2x3	red	1
1x4	red	1
1x3	red	1
1x2	red	2
1x1	red	4
inverted slope 2x2	red	8

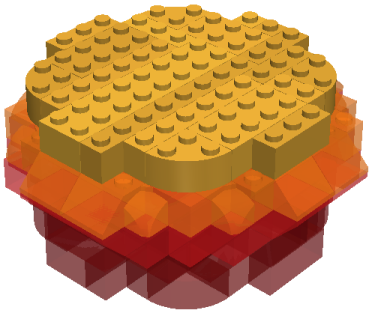
Step 4 (layer 3, z = 28.8 mm, 28 parts)

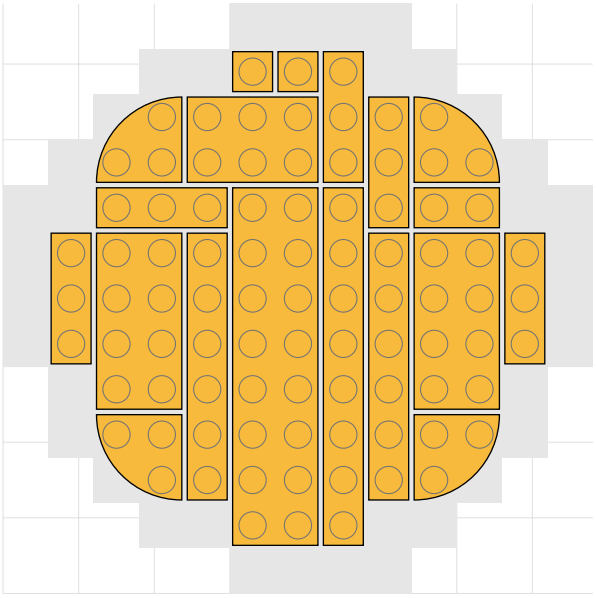




Part	Colour	Count
2x6	orange	3
1x8	orange	2
1x6	orange	1
2x3	orange	3
1x3	orange	3
1x1	orange	2
slope 2x2	orange	8
special 1x1	orange	6

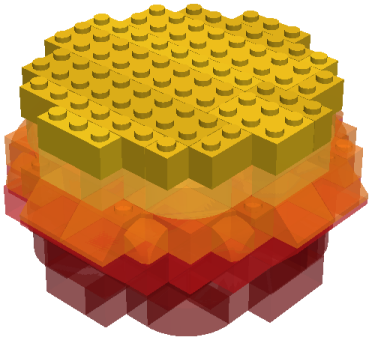
Step 5 (layer 4, z = 38.4 mm, 19 parts)

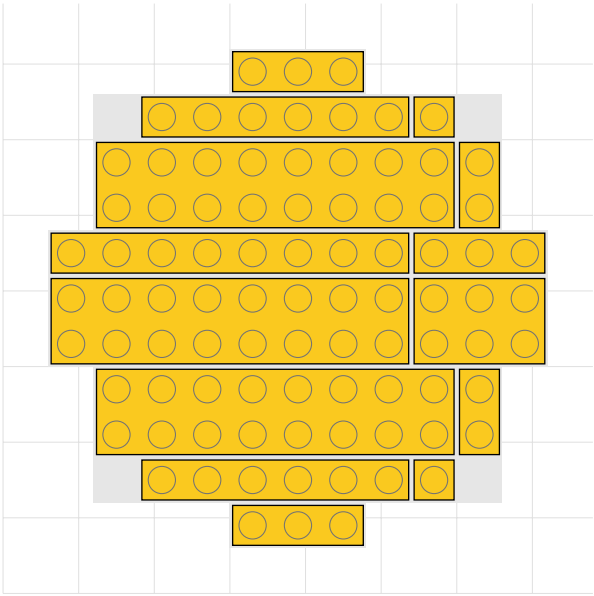




Part	Colour	Count
2x8	bright light orange	1
1x8	bright light orange	1
2x4	bright light orange	2
1x6	bright light orange	2
2x3	bright light orange	1
1x3	bright light orange	5
1x2	bright light orange	1
1x1	bright light orange	2
round corner 2x2	bright light orange	4

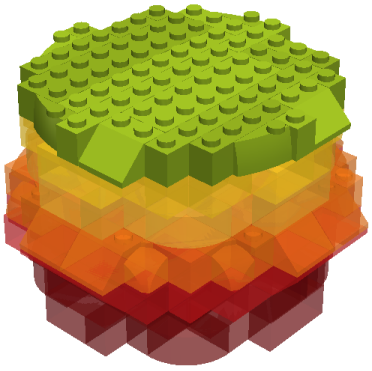
Step 6 (layer 5, z = 48 mm, 14 parts)

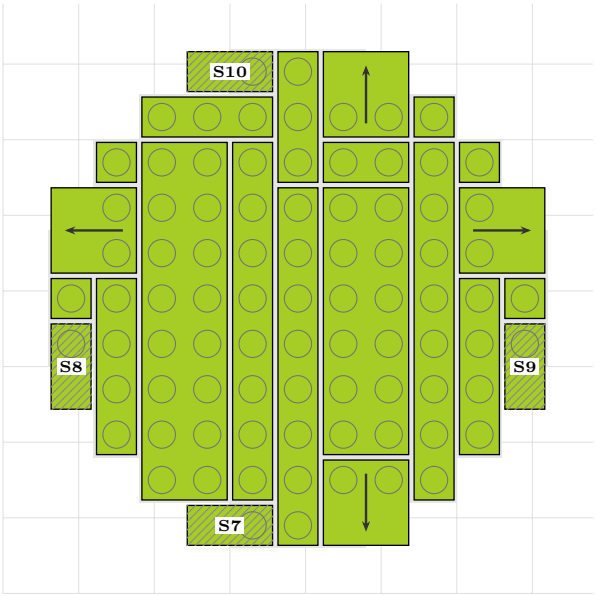




Part	Colour	Count
2x8	yellow	3
1x8	yellow	1
1x6	yellow	2
2x3	yellow	1
1x3	yellow	3
1x2	yellow	2
1x1	yellow	2

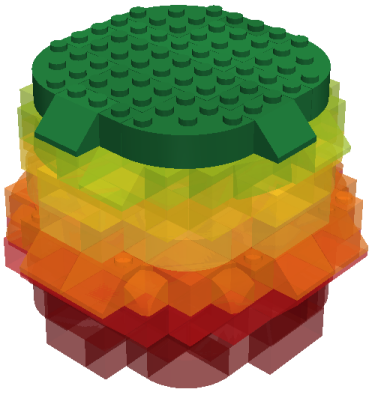
Step 7 (layer 6, z = 57.6 mm, 23 parts)

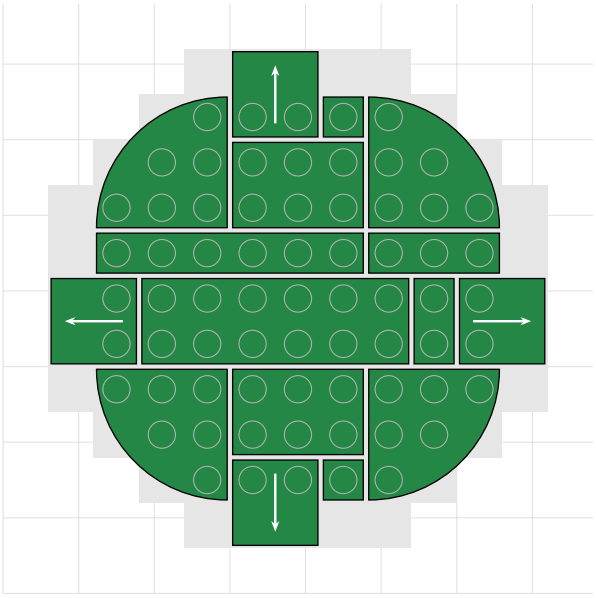




Part	Colour	Count
2x8	lime	1
2x6	lime	1
1x8	lime	3
1x4	lime	2
1x3	lime	2
1x2	lime	1
1x1	lime	5
slope 2x2	lime	4
special 1x2	lime	4

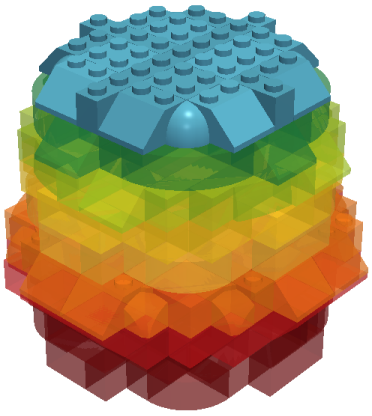
Step 8 (layer 7, z = 67.2 mm, 16 parts)

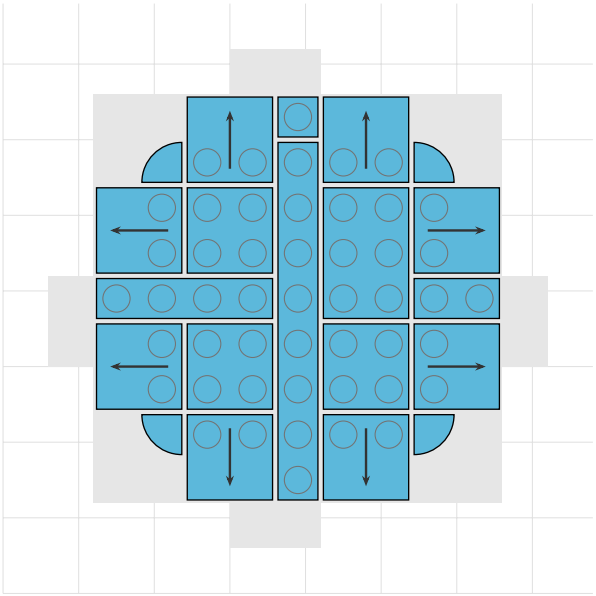




Part	Colour	Count
2x6	green	1
1x6	green	1
2x3	green	2
1x3	green	1
1x2	green	1
1x1	green	2
round corner 3x3	green	4
slope 2x2	green	4

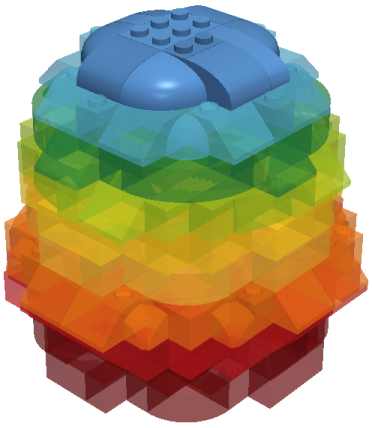
Step 9 (layer 8, z = 76.8 mm, 20 parts)

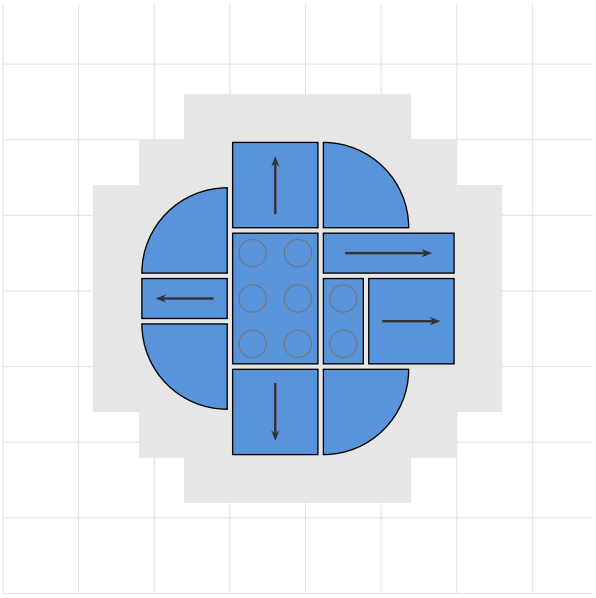




Part	Colour	Count
1x8	azure	1
2x3	azure	1
1x4	azure	1
2x2	azure	3
1x2	azure	1
1x1	azure	1
slope 2x2	azure	8
dome corner 1x1	azure	4

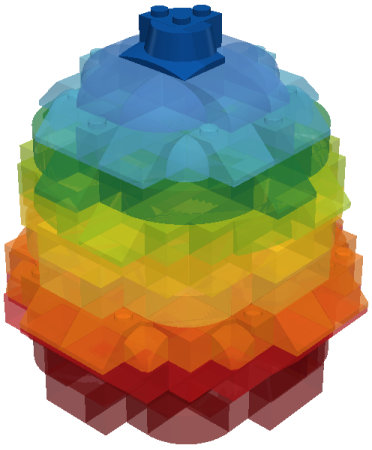
Step 10 (layer 9, z = 86.4 mm, 11 parts)

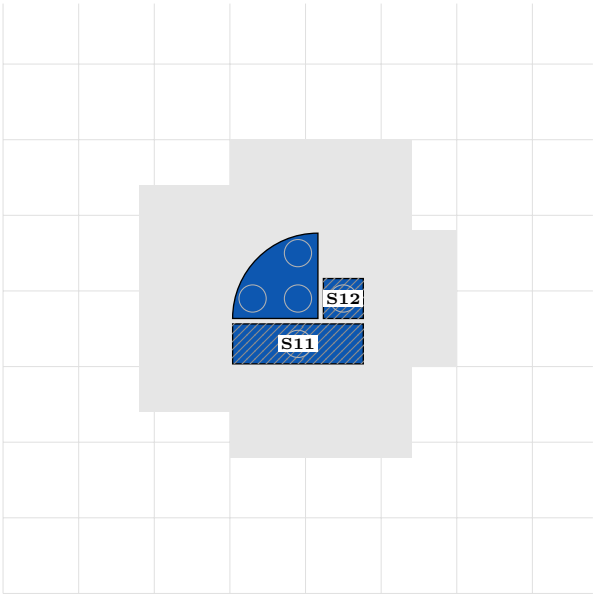




Part	Colour	Count
2x3	medium blue	1
1x2	medium blue	1
curve 2x2	medium blue	3
dome corner 2x2	medium blue	4
curve 3x1	medium blue	1
curve 2x1	medium blue	1

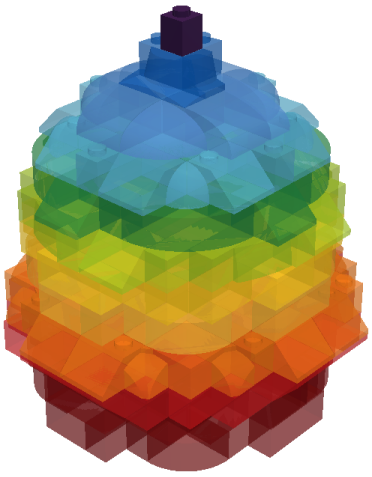
Step 11 (layer 10, z = 96 mm, 3 parts)

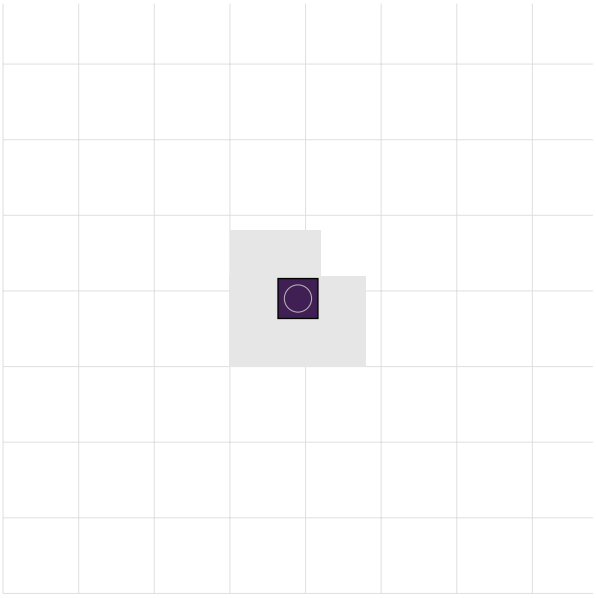




Part	Colour	Count
round corner 2x2	blue	1
special 1x3	blue	1
special 1x1	blue	1

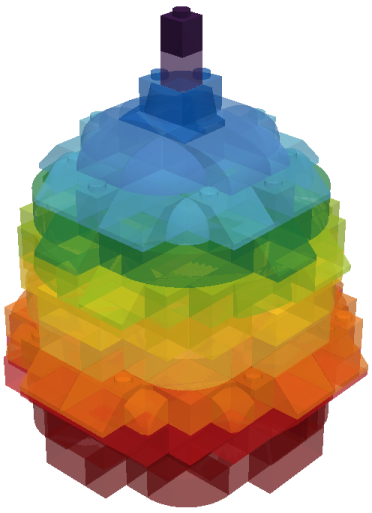
Step 12 (layer 11, z = 105.6 mm, 1 parts)

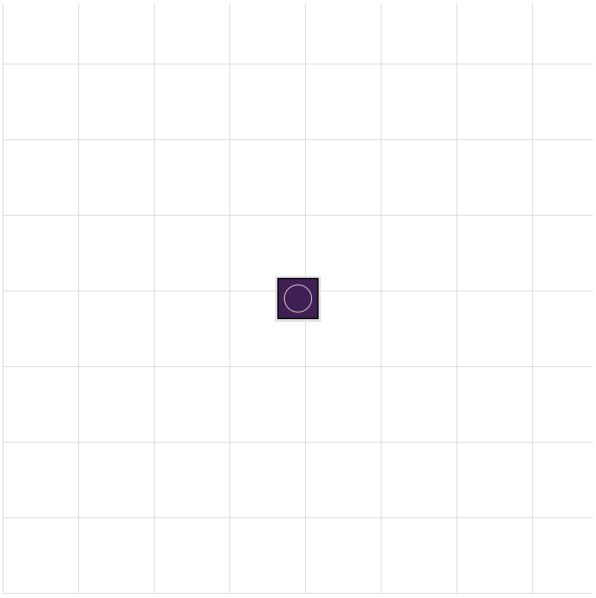




Part	Colour	Count
1x1	dark purple	1

Step 13 (layer 12, z = 115.2 mm, 1 parts)





Part	Colour	Count
1x1	dark purple	1

Special pieces

Print each from `bricks/` of the STL stack; the studs and sockets sit where it joins the bricks.

Piece	File	Layer	Cells
S1	`special_00063.stl`	3	1×1 at (1, 3)
S2	`special_00064.stl`	3	1×1 at (3, 1)
S3	`special_00065.stl`	3	1×1 at (3, 11)
S4	`special_00066.stl`	3	1×1 at (9, 1)
S5	`special_00067.stl`	3	1×1 at (9, 11)
S6	`special_00068.stl`	3	1×1 at (11, 3)
S7	`special_00120.stl`	6	2×1 at (4, 1)
S8	`special_00121.stl`	6	1×2 at (1, 4)
S9	`special_00122.stl`	6	1×2 at (11, 4)
S10	`special_00123.stl`	6	2×1 at (4, 11)
S11	`special_00187.stl`	10	3×1 at (5, 5)
S12	`special_00188.stl`	10	1×1 at (7, 6)